

Quentin Samudio

Research Engineer — Energy Modelling

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Research engineer specialised in power system modelling and energy transition analysis. Experience in large-scale modelling, urban energy systems, and collective self-consumption. Strong Python background and contributor to applied research projects on low-carbon pathways.

TECHNICAL SKILLS

Python HTML CSS
SQLite MongoDB
Supabase Git Docker
Agile React FastAPI
Flask

LANGUAGES

English C1
German A2

PERSONAL PROJECTS

dataToolBox
Python tool for multi-format data reading (DataFrame, CSV, GeoJSON, GeoPackage). github.com/CES-MDE/dataToolBox

Collective Self-Consumption Tool
Python calculation library for sizing self-consumption projects (load curves, PV, financial).

Interactive CV – Dynamic Editor
Fully editable online CV: inline editing, multi-profile (localStorage/JSON), light/dark theme, accent color picker, drag & drop, PDF auto-scaling.

INTERESTS / VOLUNTEERING

- Krav Maga – 4 years -
President, ACE (Energy Design Area) 2022–2023 -
Treasurer, T4EG (Together For Earth Grenoble) 2022–2023

PROFESSIONAL EXPERIENCE

Research Engineer — Energy Modelling 03/2024 – Present
Armines, Paris

Scientific paper (2026): first author on urban energy models and renovation strategies in the context of climate action plans (PCAET).

ADEME report: modelling and analysis for low-carbon trajectories (MAP 2050 project, CEEP MINES ParisTech).

SmartE refactoring: transformed 10 years of technical debt into a modular Python library (53,000+ lines).

Course Instructor — Advanced Python 07/2025 – 02/2026
MINES Paris – Université PSL, Paris

Designed and taught a course for around 15 engineering students.

Content: data processing, web scraping, OOP, best practices, Git/GitHub.

Intern — Collective Self-Consumption Project Manager 05/2023 – 12/2023
Enogrid, Paris

Supported 12 clients on collective self-consumption projects.

Developed a Python module for project sizing (50+ participants), integrated into Enolab.

Intern — Assistant Engineer 05/2022 – 07/2022
Buildings Department, City Hall of Saint-Denis

Analysis of 200+ energy audits (Tertiary Decree).

Site supervision for health and early childhood facilities.

Student Associate — Fab Manager 11/2021 – 05/2021
Fablab Green-er, ENSE3, Grenoble

Managed and trained users on fablab equipment.

Organised DIY wind turbine workshops.

EDUCATION

Master of Engineering — Energy Systems and Markets
ENSE3 – Grenoble INP, 01/2020 – 01/2024

Architecture Certificate – PARIN Program (Architecture-Engineering)
ENSAG, Grenoble, 01/2023

Graduate Program — Environment & Natural Resources (Erasmus)
University of Reykjavik, Iceland, 01/2022

Preparatory Classes PSI* (Physics and Engineering Sciences)
Lycée Jacques Decour, Paris, 01/2017 – 01/2020